

GEN

Please review the content of all the cells carefully.

Weeks	Steps	Task description	Before your tutorial	In-tutorial	After your tutorial	Documents that will be marked
<u>Week 5</u>	1	Forming Groups		✓		Submit by Sunday of Week 5 at 11:59 PM): • Problem statement
	2	MS Teams		✓		

2026 T1 COMP9511 Assignment 2

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	3	Writing the Problem Statement and Product Statement		✓	✓	• Product description statement
<u>Week 7</u>	4	Developing a Questionnaire	✓			Submit by Sunday of Week 7 at 11:59 PM): • Consent documentation • Questionnaire interview questions • Analysis results of the interview
	5	Run a Pilot Interview during the tutorial time		✓		
	6	Actual Interviews			✓	
	7	Analysing the Interviews			✓	
<u>Week 8</u>	8	Users Personas and Scenarios	✓	✓	✓	Submit by Sunday of Week 8 at 11:59 PM): • Personas • Context Scenarios • Requirements list
	9	Functional and Non-Functional Requirements	✓	✓	✓	
<u>Week 9</u>	10	Prototype	✓			Submit in-tutorial: • Prototype • A usability test plan • Pre-test and post-test questionnaires for your usability test • Running a usability test in your tutorial.
	11	Preparing the Usability Test Plan	✓			
	12	Running a Usability Test		✓		
<u>Week 10</u>	13	Running your own Usability Tests	✓			Submit in-tutorial: • Live presentation • The final prototype • Aggregated issue table of all the usability tests
	14	Compiling the Usability Testing Outcomes in an Issues Table	✓			
	15	Revise the Design	✓			
	16	Live Presentation		✓		
	17	Peer Review			✓	

Problem Statement:

 Problem Statement

Problem Statement

Problem Statement

The Current online shopping websites present users with attractive user interfaces and Helpers of convenient shopping experience; however, they do not tend to support the user sufficiently in order to resolve the problem. These issues include inability to clearly explain refunds, give advice on consumer rights, and make the process of complaining complex, which confuse and frustrate the users. The transparency and user support are not a priority on many platforms, and the users can hardly know their rights in case of a broken product, late delivery time, or false product descriptions.

Moreover, other websites of e-commerce are not necessarily accessible and user-friendly. Old users and users with visual impairments find it very difficult to use dense layouts, small fonts, low contrast, and complex navigation systems. These problems make usability lower and pose a challenge to a large group of users who could need a more simplified and supportive interface.

To overcome these issues, this project suggests the example of developing an e-commerce web application where consumers rights awareness and support are provided directly in the shopping process. The platform is inspired by the current trends in UI/UX design, including card layouts, product carousels, and the interactive elements and, most importantly, improves the user experience, adding such features as definite policies of returns, easy access to help, and the simplification of complaint forms. It is to develop a system that does not merely assist in doing shopping but also enables users to acquire the knowledge and means to learn to utilize their rights as consumers in an accessible and user-friendly manner.

Product Description

The offered system is the contemporary e-commerce web application, which is a synthesis of online shopping with the support of consumer rights. Users get to view products in an easy to use visually stimulating interface and find an explicit reply to refunds, returns, and complaints. The site has features of the design that are accessibility oriented like large text size, straightforward navigation, and contrast designs in order to enable aged users. The system is expected to achieve transparency, efficiency and ease of use by incorporating consumer protection functionalities to a familiar shopping interface.

week7

This week output:

1. final Questionnaire
2. Questionnaire result
3. Consent documentation

1 .Questionnaire

✓ 1. Questionnaire

♦ General Shopping Behaviour

1. How often do you shop online?

- Daily
- Weekly 1
- Monthly
- Rarely

2. Which platforms do you usually use?

- Amazon 1
- eBay
- Temu
- Other: _____

3. What is most important when shopping online? (Select up to 2)

- Price 1
 - Product quality 1
 - Reviews
 - Easy returns
 - Fast delivery
-

♦ Problems & Pain Points

4. Have you ever faced issues while shopping online?

- Yes
- No 1

5. If yes, what kind of issues did you face?

- Faulty product
- Late delivery
- Difficult returns
- No refund
- Poor customer support

6. How easy is it to find refund/return information on shopping websites?

(Scale 1–5) 3

1 = Very difficult, 5 = Very easy

7. Have you ever tried to make a complaint on a shopping platform?

- Yes 1
- No

8. If yes, how was your experience? (Short answer)

interface difficult to use

♦ Needs & Expectations

9. What features would improve your shopping experience? (Select any)

- Clear return/refund info
- Easy complaint system
- Order tracking
- Simple navigation 1
- Better product info 1

10. How important is it to clearly understand your rights (refunds, returns)?
(Scale 1–5) 5

11. Would you use a feature that helps you report problems easily?

- Yes
 - Maybe 1
 - No
-

♦ Accessibility (VERY IMPORTANT FOR MARKS)

12. Do you find some websites difficult to read or navigate?

- Yes 1
- No

13. What problems do you face?

- Small text
- Too much clutter
- Poor contrast 1
- Confusing layout

14. What would improve usability for you? (Open-ended)

clear contrast

♦ Open Feedback

15. What would your ideal shopping website look like?

clear navigation , easy to search function and product.

2 .Questionnaire

✓ 1. Questionnaire

♦ General Shopping Behaviour

1. How often do you shop online?

- Daily
- Weekly 1
- Monthly
- Rarely

2. Which platforms do you usually use?

- Amazon 1
- eBay
- Temu
- Other: _____

3. What is most important when shopping online? (Select up to 2)

- Price
 - Product quality 1
 - Reviews
 - Easy returns
 - Fast delivery
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♦ Problems & Pain Points

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(Scale 1–5)

1 = Very difficult, 5 = Very easy 4

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- Yes 1
- No

8. If yes, how was your experience? (Short answer)

Difficult to return

♦ Needs & Expectations

9. What features would improve your shopping experience? (Select any)

- Clear return/refund info 1
- Easy complaint system
- Order tracking
- Simple navigation
- Better product info

10. How important is it to clearly understand your rights (refunds, returns)?
(Scale 1–5) 4

11. Would you use a feature that helps you report problems easily?

- Yes
 - Maybe 1
 - No
-

♦ Accessibility (VERY IMPORTANT FOR MARKS)

12. Do you find some websites difficult to read or navigate?

- Yes 1
- No

13. What problems do you face?

- Small text
- Too much clutter
- Poor contrast
- Confusing layout 1

14. What would improve usability for you? (Open-ended)

easy to return

♦ Open Feedback

15. What would your ideal shopping website look like?

easy to return. fluent navigation

analysis results

Week 7 Interview / Survey Analysis Summary

Project Title: ShopCart User Research

Number of Responses: 14

1. Overview

This user research was conducted to investigate users' online shopping experiences, with a particular focus on consumer rights, complaint handling, return and refund processes, and usability issues in e-commerce platforms. The questionnaire also explored users' expectations for a more accessible and user-friendly shopping system. According to the questionnaire document, the study aimed to improve the design of an e-commerce platform called **ShopCart**, and all data was collected for academic purposes only.

The findings suggest that online shoppers frequently encounter problems during purchasing and post-purchase stages, especially in relation to delayed delivery, difficult returns, unclear refund information, and inefficient complaint procedures. These results indicate a clear need for a platform that not only supports shopping tasks, but also helps users understand and exercise their consumer rights. This aligns well with the assignment theme of **legal information or consumer rights systems**.

2. Participant Profile

A total of **14 participants** completed the questionnaire. Based on the analytics screenshots you shared, all respondents agreed to participate in the study for academic purposes.

The age distribution shows that the majority of respondents were **18–25 years old (57.1%)**, while the remaining participants were distributed across **Under 18, 26–35, and 36–50** age groups, each representing **14.3%** of the sample. There were no respondents aged 50+ in this round of data collection.

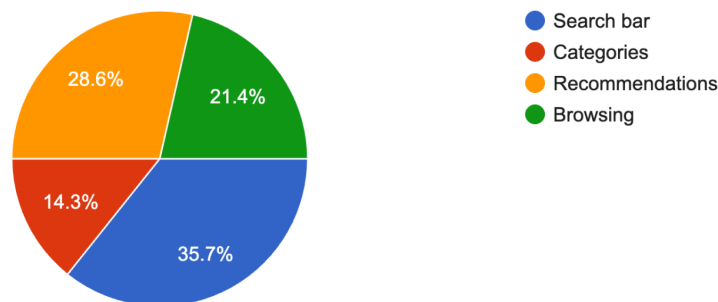
In terms of shopping frequency, most participants were regular online shoppers. **42.9%** reported shopping online weekly, **28.6%** monthly, **21.4%** daily, and only **7.1%** said they shopped online rarely. This suggests that the respondents had sufficient experience with online shopping platforms to provide meaningful and relevant feedback.

3. Current Online Shopping Behaviour

The most commonly used shopping platforms were **Amazon (57.1%)** and **Temu (50.0%)**, followed by **eBay (21.4%)**. A few additional platforms were mentioned in the open-response entries, such as Flipkart, AliExpress, and Taobao. This indicates that respondents were familiar with a range of online shopping environments rather than relying on a single service.

When asked how they usually find products, the most common answer was **search bar (35.7%)**, followed by **recommendations (28.6%)**, **browsing (21.4%)**, and **categories (14.3%)**. This pattern suggests that efficient search functionality and clear information architecture are important elements of the user experience.

How do you usually find products?
14 responses



4. What Users Value Most When Shopping

The ranking question indicates that **price** and **product quality** were the two most important considerations for most respondents. Product reviews also appeared to be valued, although generally not as highly as price and quality. These findings suggest that users care not only about affordability, but also about trustworthiness and product reliability.

Navigation was also shown to be highly important. In the questionnaire, users were asked how important ease of navigation is on a website. The analytics indicate that most participants rated this factor very highly, with the majority selecting **4 or 5 out of 5**. This shows that clear navigation is a major determinant of overall satisfaction in online shopping.

5. Common Problems Faced by Users

One of the strongest findings from the survey is that **85.7%** of respondents had experienced problems while shopping online. Only **14.3%** stated that they had never encountered such issues. This demonstrates that online shopping problems are not isolated incidents, but rather common experiences among users.

The most frequently reported issues were:

- **Late delivery (57.1%)**
- **Difficult returns (42.9%)**
- **Faulty products (28.6%)**
- **Poor customer support (28.6%)**
- **No refund (21.4%)**

These results show that users' frustrations often occur after the purchase has already been made. In other words, the most critical pain points are not limited to product discovery or checkout, but extend into returns, refunds, support, and complaint resolution.

6. Return, Refund, and Complaint Experience

The responses suggest that return and refund information is not consistently easy to find. Most users rated this experience at the middle level, indicating that the information is available in some cases, but often not presented clearly enough.

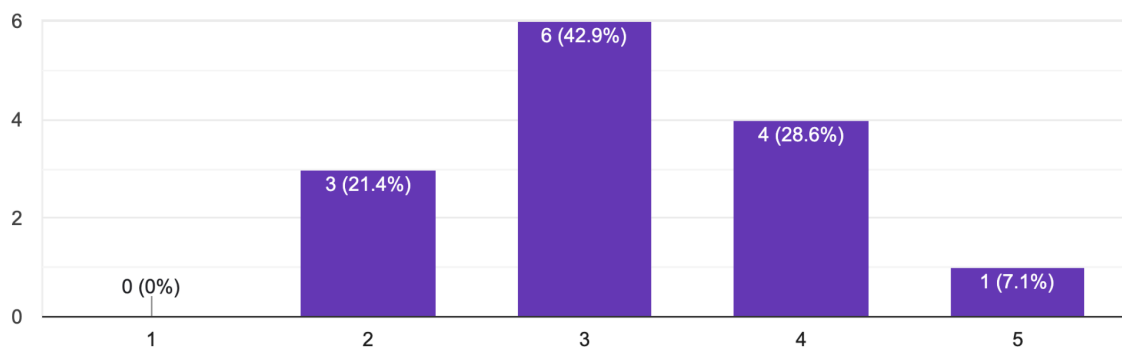
A significant proportion of respondents, **71.4%**, said they had tried to make a complaint before. This shows that complaint handling is a highly relevant user need rather than an uncommon edge case.

However, users' evaluations of the complaint process were mixed or negative. A large proportion described the process as neutral, difficult, or frustrating. The main reasons given for difficulty were:

- **Too many steps (42.9%)**
- **No clear instructions (28.6%)**
- **Slow response (14.3%)**

How easy is it to find return/refund information?

14 responses



These findings suggest that existing complaint systems are often overly complicated and poorly guided. Users do not simply need a complaint feature; they need a complaint process that is visible, easy to follow, and efficient.

7. Awareness of Consumer Rights

The questionnaire also explored how important it is for users to clearly understand their rights regarding refunds and returns. The open-ended responses consistently showed that this issue matters greatly to users. Many participants emphasised that they want to know

what they can do before buying, especially if something goes wrong later. Others noted that many websites do not explain return and refund rules clearly enough.

Several responses also implied that clear consumer-rights information would be especially helpful for older users or for those who need more straightforward guidance. This is important because the assignment specifically requires students to consider the needs of **elderly users with visual impairments** when developing their system.

Overall, the data suggests that users want platforms to provide more transparent explanations of policies, procedures, and rights. This reinforces the idea that the proposed system should function not only as a shopping interface, but also as a consumer-rights support platform.

8. Features Users Want

Participants were asked which features would improve their experience. The most frequently selected features were:

- **Clear return/refund information (71.4%)**
- **Order tracking (57.1%)**
- **Better product details (42.9%)**
- **Easy complaint system (35.7%)**
- **Simple navigation (28.6%)**

These responses highlight a strong preference for clarity, transparency, and support. Users want to be informed, guided, and reassured throughout the shopping process.

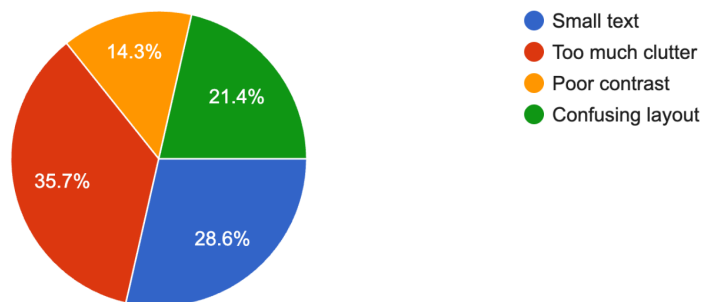
The responses to the “Report Issue” question further support this point. **57.1%** of participants said they would use such a feature, while **14.3%** said maybe. This suggests that a dedicated issue-reporting function would be useful for a majority of users, especially if it is integrated with clear support pathways.

9. Usability and Accessibility Findings

Another important theme in the data concerns usability. **78.6%** of respondents said that they find some websites difficult to use. The most common interface problems they identified were:

- **Too much clutter (35.7%)**
- **Small text (28.6%)**
- **Confusing layout (21.4%)**
- **Poor contrast (14.3%)**

What problems do you face?
14 responses



•
When asked what would improve usability, users most frequently chose:

- **Easy navigation (57.1%)**
- **Clear buttons (50.0%)**
- **Faster loading (50.0%)**
- **Clean layout (42.9%)**
- **Easy returns (35.7%)**
- **Better contrast (28.6%)**
- **Visible support (28.6%)**
- **Bigger text (21.4%)**

These findings are particularly valuable because they point directly to accessibility-related design improvements. The assignment brief states that the interface should consider the needs of elderly users with visual impairments. The survey results support this requirement by showing that users already struggle with small text, cluttered pages, confusing layouts, and low-clarity interfaces.

10. Themes from Open-Ended Responses

The open-ended responses reveal several recurring themes.

First, many participants described an ideal shopping website as **simple, clean, and easy to use**. They wanted a design that allows them to complete tasks quickly without unnecessary complexity.

Second, users frequently referred to the importance of **clear information**, especially regarding returns, refunds, and complaint processes. They did not want important policies to be hidden or difficult to interpret.

Third, participants repeatedly mentioned the need for **readability**, including larger text, more understandable wording, and a clearer page structure.

Finally, several responses emphasised the importance of **efficient support**, such as a complaint process that takes only one or two steps, honest product descriptions, and visible after-sales assistance.

Taken together, these responses suggest that users value transparency, simplicity, and confidence in the platform more than decorative or advanced features.

11. Key Findings

Based on the survey results, the main findings are as follows:

1. Most users shop online regularly and are therefore familiar with common platform issues.
2. Online shopping problems are widespread, especially in delivery, returns, refunds, and support.
3. Users strongly value clear navigation, product quality, and price.
4. Complaint procedures are often seen as too complex and unclear.
5. Users want clearer information about their consumer rights, especially regarding returns and refunds.
6. Simplicity, readability, and accessibility are essential for a better user experience.
7. Features such as issue reporting, visible support, order tracking, and clear refund information would significantly improve usability.

12. Design Implications

These findings can directly inform the next stages of the design process, including personas, scenarios, and requirements. According to the assignment brief, Week 8 requires students to produce personas, scenarios, and a requirements list based on the interview results.

The results suggest that the proposed system should include:

- clear and accessible refund/return information
- a simplified complaint or issue-reporting process
- visible support channels
- straightforward navigation and search
- larger text and stronger visual clarity
- accessible layouts suitable for users with visual difficulties

These insights can also support the development of personas such as a frequent online shopper, a user who has experienced refund difficulties, or an older user who needs a more readable and guided interface.

13. Limitations

Although the survey provided useful insights, it also has several limitations.

First, the sample size was relatively small, with only **14 responses**. Second, the majority of participants were young adults aged 18–25, so the views of older users were underrepresented. Third, some open-response answers were brief or incomplete, which limited the depth of analysis. Finally, because the study used questionnaire data rather than fully developed interviews, some responses were less detailed than they might have been in a face-to-face discussion.

Despite these limitations, the findings still reveal clear trends that are useful for the next stage of the project.

14. Conclusion

In conclusion, the survey results show that online shoppers commonly face challenges related to late delivery, difficult returns, poor support, and unclear complaint procedures. Users want platforms that are not only efficient for shopping, but also transparent, easy to navigate, and supportive when problems occur. They also place strong value on understanding their consumer rights and being able to act on them easily.

Therefore, the future design of ShopCart should prioritise **clarity, accessibility, complaint support, and consumer-rights information**. These priorities are supported both by the quantitative results and by the repeated themes in the open-ended responses. The findings provide a solid basis for developing personas, scenarios, and system requirements in the next phase of the assignment.

Consent documentation

ShopCart User Research – Consent Form

We are conducting this research as part of the Human Computer Interaction (COMP9511) course at the University of New South Wales.

This study aims to understand users' online shopping experiences and improve the design of an e-commerce platform called ShopCart.

Your participation is voluntary, and you may withdraw at any time. All responses will remain confidential and will be used only for academic purposes.

Your privacy, observers & recording of data

Please be assured that the information and opinions you provide today will be used only for research purposes.

We will record information about how you interact with the website through questionnaires and notes. This will help us understand how to improve usability and design. All collected data will remain anonymous.

Your name will appear only on this form and will be kept separately for consent verification. This form will be destroyed after the completion of the Human Computer Interaction course.

The data recorded may be used for:

- Analysis of user interactions
- Documenting findings for the assignment
- Providing feedback to improve the system design
- Teaching or training purposes (demonstrating research process)
- Assessment by UNSW teaching staff

The data will not be shared with any external parties without your consent.

* Indicates required question

final Questionnaire

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The data will not be shared with any external parties without your consent.

...

Please confirm that you understand and agree to the following: *

- ☐ I understand that this research is for academic purposes only
- ☐ I understand my responses will remain confidential
- ☐ I understand I can withdraw at any time
- ☐ I agree to participate in this study

Name (optional):

Date:

Month, day, year



Do you agree to participate in this study for academic purposes?

- ☐ Yes
- ☐ No

Q1

Do you agree to participate in this study?

- Yes
- No

Q2

What is your age group?

- Under 18
- 18–25
- 26–35
- 36–50
- 50+

Q3

How often do you shop online?

- Daily
- Weekly
- Monthly
- Rarely

Q4

Which platforms do you use? (*Select all that apply*)

- Amazon
- eBay
- Temu
- Flipkart
- Other

Q5

Rank what matters most when shopping (1 = most important)

Factor	1	2	3	4	5
Price	☐	☐	☐	☐	☐
Product quality	☐	☐	☐	☐	☐
Reviews	☐	☐	☐	☐	☐
Easy returns	☐	☐	☐	☐	☐
Fast delivery	☐	☐	☐	☐	☐

Q6

How important is easy navigation?
(1 = Not important, 5 = Very important)

	1	2	3	4	5	
Not important	☐	☐	☐	☐	☐	Very important

Q7

How do you usually find products?

- Search bar
- Categories
- Recommendations
- Browsing
- Other: _____

Q8

Have you faced problems while shopping online?

- Yes
- No

Q9

What issues have you faced? *(Select all that apply)*

- Faulty product
- Late delivery
- Difficult returns
- No refund
- Poor customer support

Q10

How easy is it to find return/refund information?

	1	2	3	4	5	
Very difficult	☐	☐	☐	☐	☐	Very easy

Q11

Have you made a complaint before?

- Yes
- No

Q12

How was your complaint experience?

- 😊 Very easy
- 🙂 Easy
- 😐 Neutral
- 😞 Difficult
- 😡 Very frustrating

Q13

What made it difficult?

- Hard to find complaint option
- Too many steps
- No clear instructions
- Slow response
- Other: _____

Q14

Which features would improve your experience?

- Clear refund/return info
- Easy complaint system
- Order tracking
- Simple navigation
- Better product details
- Other: _____

Q15

How important is it to clearly understand your rights (refunds, returns)?

Q16

Would you use a “Report Issue” feature?

- Yes
- Maybe
- No

Q17

Do you find websites difficult to use?

- Yes
- No

Q18

What issues do you face?

- Small text
- Cluttered layout
- Poor contrast
- Confusing navigation

Q19

What would improve usability?

- Bigger text
- Clean layout
- Easy navigation
- Clear buttons
- Better contrast
- Faster loading
- Easy returns
- Visible support

Q20 Describe your ideal shopping website in your own words:



Thank you for your time and participation.
Your responses will help improve the design of the ShopCart platform.



1

2

3

4

5



After section 2 Submit form



Section 3 of 3

Survey End



"Thank you. Since you did not consent, the survey has ended."

week8

This week output:

Personas: 3–4 persona

Context Scenarios: 3–4 scenario

Requirements list: 1–2 pages

Personas

Week 8 Draft

1. Personas

Persona 1 — Frequent Young Online Shopper

Name: Emily Chen

Age: 22

Occupation: University student

Background: Emily shops online every week for clothes, electronics, and daily items. She is comfortable using large shopping platforms and usually finds products through the search bar or recommendations.

Goals:

- Find products quickly
- Compare price, quality, and reviews
- Get clear information about refunds and returns before buying
- Resolve issues efficiently if something goes wrong

Frustrations:

- Return/refund policies are often hard to find
- Complaint processes are too long and confusing
- Customer support is slow or unhelpful
- Important information is buried in cluttered pages

Needs:

- Clear navigation and search
- Easy-to-read refund and return information
- A simple complaint/report issue flow
- Order tracking and visible support options

Quote:

“I want to know my rights before I buy, not after I run into a problem.”

Persona 2 — Careful Value-Seeking Shopper

Name: Daniel Wu

Age: 31

Occupation: Office worker

Background: Daniel shops online monthly and pays close attention to price, product quality, and reviews. He wants trustworthy product information and reliable after-sales service.

Goals:

- Make informed purchase decisions
- Avoid faulty products or misleading listings
- Track orders easily
- Get a refund or return without unnecessary effort

Frustrations:

- Product details are sometimes incomplete
- Delivery delays are common
- It is hard to know what action to take when a seller does not cooperate
- Complaint channels are not obvious

Needs:

- Better product details and trustworthy reviews
- Clear steps for complaint escalation
- Transparent consumer-rights guidance
- Fast and visible support

Quote:

“If the process is fair and clear, I’m much more likely to trust the platform.”

Persona 3 — User with Refund Difficulty Experience

Name: Sarah Nguyen

Age: 27

Occupation: Retail assistant

Background: Sarah has previously received a faulty product and struggled to get a refund. She now feels anxious when buying online, especially from unfamiliar platforms.

Goals:

- Understand what she can do if something goes wrong
- Report issues quickly
- Receive updates during complaint handling
- Feel reassured during post-purchase support

Frustrations:

- No clear instructions for making a complaint
- Too many steps in return/refund requests
- Lack of visibility on complaint progress
- Fear of losing money when the process is unclear

Needs:

- Report Issue button
- Step-by-step complaint guidance
- Complaint status tracking
- Plain-language consumer-rights explanations

Quote:

"I don't mind reporting a problem, but I need the process to be simple and visible."

Persona 4 — Older User with Visual Impairment Needs

Name: Margaret Lee

Age: 62

Occupation: Retired teacher

Background: Margaret shops online occasionally. She finds some websites difficult to use because of small text, cluttered layouts, and low contrast. She prefers straightforward wording and larger clickable elements.

Goals:

- Read information comfortably
- Understand return/refund rights clearly
- Navigate the site without confusion
- Get support without needing to search too much

Frustrations:

- Small text and crowded pages
- Poor contrast between text and background
- Confusing page structure and hidden support options
- Legal or policy language that is hard to understand

Needs:

- Large text options
- Strong contrast and clean layout
- Clear buttons and labels
- Easy-to-follow explanations of rights and procedures

Quote:

"I need the website to guide me clearly, not make me work hard to understand it."

2. Context Scenarios

Scenario 1 — Checking refund rights before purchase

Emily wants to buy headphones from ShopCart. Before placing the order, she wants to know what happens if the item is faulty or different from the description. She visits the product page and expects to quickly find a clear summary of return and refund rights. She wants the platform to explain the rules in simple language so she can feel confident before completing the purchase.

Scenario 2 — Handling a delayed order

Daniel ordered a household item that was supposed to arrive three days ago. He logs into ShopCart to check the status of the order and understand what he can do next. He wants to see order tracking information, access support easily, and know whether he is entitled to compensation, cancellation, or another action if the delay continues.

Scenario 3 — Reporting a faulty product

Sarah receives a damaged item and wants to report the issue immediately. She opens ShopCart and looks for a clear “Report Issue” or complaint option. She expects the platform to guide her through a short, step-by-step process, explain what evidence is needed, and show what outcomes are possible, such as replacement, refund, or escalation.

Scenario 4 — Older user seeking readable rights information

Margaret wants to buy a kitchen appliance online, but she is worried about whether she could return it if it does not work properly. She visits ShopCart and needs a simple, readable interface with large text, strong contrast, and clear buttons. She wants consumer-rights information written in plain language, and she needs support options that are easy to notice and use.

3. Requirements List

Functional Requirements

1. The system must provide clear return and refund information on product pages and support pages.
2. The system must allow users to report an issue through a dedicated complaint/reporting feature.
3. The system must provide step-by-step guidance for return, refund, and complaint processes.
4. The system must allow users to track order status and complaint status.
5. The system must provide visible support channels such as contact form, chat, or help centre access.
6. The system must provide clear product details, including delivery information, seller information, and policy summaries.
7. The system must include search and simple navigation to help users find products and rights-related information quickly.
8. The system must explain consumer rights related to refunds, returns, and complaints in plain language.
9. The system must show users what actions are available when common problems occur, such as late delivery, faulty product, or no refund.
10. The system should save users' past issue reports and support requests in their account dashboard.

Non-Functional Requirements

1. The system must be easy to navigate, with a clear and consistent layout.
 2. The interface must use readable text sizes and allow larger text where needed.
 3. The interface must provide sufficient contrast between text and background.
 4. The interface must minimise clutter and avoid overwhelming users with too much information on one screen.
 5. Buttons, links, and key actions must be clearly labelled and easy to identify.
 6. Information about returns, refunds, and complaints must be easy to find within a small number of steps.
 7. The system must support accessibility needs, especially for elderly users with visual impairments.
 8. The platform should use plain, understandable language instead of complex legal wording where possible.
 9. Key pages should load quickly and respond efficiently.
 10. The design should build user trust through transparency, visible support, and honest information presentation.
-

Context Scenarios

Requirements list

kap

Week 8: Personas, Scenarios & Requirements

User Personas

We've created some personas based on what people told us in interviews and surveys. The main problems that came up were that it's hard to return things, the complaint system is confusing, the websites are cluttered, and some people have trouble accessing them because the text is too small or the layout is bad.

Persona 1

Name: Alex Chen

Age: 21

Occupation: University Student

Tech Level: High

Shopping Frequency: Daily/Weekly

Goals

- Quickly find products
- Complete purchases efficiently
- Easily resolve issues

Frustrations

- Too many steps in checkout/complaints
- Difficult return process
- Slow response from support

Needs

- Fast navigation (seen in your homepage categories)
- Clear refund/return system (missing in UI → important insight)
- Minimal steps

Scenario 1

Alex visits the platform to quickly buy a product using the search or category section. After purchasing, the product turns out to be faulty. Alex tries to request a return but expects the system to provide a clear, simple process with minimal steps and quick resolution.

Persona 2

Name: Sarah Lee

Age: 30

Occupation: Working Professional

Goals

- Reliable shopping
- Clear policies before purchase

Frustrations

Hidden return/refund information

- Poor complaint handling

Needs

- Transparent product + policy info (partially shown in your product page)
- Strong customer support visibility

Scenario 2

Sarah explores a product page and checks details before purchasing. After encountering an issue, she wants a clear complaint system and expects transparency in refund policies.

Persona 3

Name: John Smith

Age: 60+

Tech Level: Low

Goals

- Easy-to-use system

- Readable content

Frustrations

- Small text in product cards
- Dense layout in product grid
- Too many options

Needs

- Larger fonts
- Clean layout
- Clear buttons

Scenario 3

John browses the homepage but struggles to read product details due to small text and crowded layout. He wants a simpler interface that clearly guides him through selecting and purchasing products.

Persona 4

Name: David Wong

Age: 25

Goals

- Quickly report issues
- Track complaints

Frustrations

- “Too many steps”
- “No clear instructions”
- “Slow response”

Needs

- One-click complaint/report feature
- Clear step-by-step guidance

Scenario 4

David receives a defective item and wants to report the issue. However, the system does not clearly show complaint options. He expects a simple and direct reporting feature.

Requirements

Functional Requirements

1. The system must allow users to search for products
2. The system must allow category-based browsing
3. The system must display product details
4. The system must allow users to add items to cart
5. The system must allow checkout and payment
6. The system must provide order summary
7. The system must allow users to track orders
8. The system must provide return/refund request functionality
9. The system must allow users to submit complaints
10. The system must provide customer support access

Non-Functional Requirements

Usability

- Interface must be simple and intuitive
- Reduce number of steps in key tasks
- Clear navigation structure

Accessibility

- Provide larger text options
- High contrast colours
- Reduce clutter in product grids
- Clear button labels

Performance

- Pages must load quickly
- Smooth transitions (carousel, etc.)

Security

- Secure payment methods (shown in your UI)
- Protect user data

Reliability

- System must function without crashes

Learnability

- First-time users should easily understand how to use system

ANALYSIS

The findings from the survey and interface evaluation indicate that users struggle most with returns, complaint processes, and usability issues such as cluttered layouts and small text. These insights directly influenced the system requirements and highlight the need for a more transparent, accessible, and user-friendly e-commerce platform.

week9

“Consent documentation”的副本

Consent form for participation in ShopCart User Research

Please read the following information carefully.

Student Kapil Vinayak is conducting today's research on behalf of The University of New South Wales. The research is being used as a class assessment exercise for a course Human Computer Interaction (COMP3511) at The University of New South Wales. The topic is concerned with the design of *e-commerce web application*.

As a participant in the session, we will ask you to help us understand how you work and your wants and needs for e-commerce web application. The observations and information we collect today will help us to build improvements into the designs and will be used as part of our University assignment work that help us to learn about involving users in the design process. If you have any questions, we encourage you to ask them.

As your participation is entirely voluntary, you may choose to withdraw from the session at any time.

If you have any further queries you can contact Student Kapil Vinayak on Kapil10@gmail.com.

Your privacy, observers & recording of data

Please be assured that the information and opinions you provide today will be used only for research purposes.

We will record information about how you might use and interact with the site (including on videotape, audiotape and questionnaire). This will help us see how to make the design more useful and easier to use. For any and all uses of this recorded data, we undertake to guarantee your anonymity. Your name will appear only on this form, which will be filed separately, and is kept only to verify your consent to participation in this session. The form will be destroyed after the completion of the Human Computer Interaction course.

The data recorded may be used for:

- Analysis, to review interactions after the session
- Documenting the findings of the session
- Providing feedback to the project team about the design and how well it does/does not meet peoples' needs or communicating the same to other stakeholders within the organisation sponsoring the research
- Teaching or training purposes by group members but only for demonstrating the process of a typical research session (eg. how to run this type of session).

- Notes and questionnaires may be passed on to teaching staff of the University for course assessment purposes. If you sign this sheet, you give your permission to have notes recorded and for the notes to be passed on to course teaching staff.
- The notes will not be transferred to anyone else unless your consent has been obtained.

Please continue over...

Non-disclosure

By participating in this session, you may be privy to information that the course regards as confidential, and must accept that the participant is the sole and exclusive owner of this confidential information.

You must not at any time after the session disclose to any other person or use, adopt, be a party to, aid or abet any disclosure, use, adoption or employment of this confidential information either directly or indirectly for any purpose.

If you have read the above information carefully and understand and consent to its contents, please complete and sign the following statement.

I,

(Participant's Name – please print)

have read the above passage carefully and understand its contents. I voluntarily agree to participate in the e-commerce web application. evaluation conducted by Jiajun Tang.

(Participant's Signature)

(Date)

(Researcher's
Signature)

(Date)

Usability Test Plan

Usability Test Plan

Project Title

Usability Test Plan for ShopCart Prototype

1. Purpose

This usability test aims to evaluate the usability of the **ShopCart prototype**, an e-commerce interface that supports browsing, purchasing, returning products, and reporting order issues. The test focuses on whether users can successfully complete common shopping and after-sales tasks efficiently, confidently, and with minimal confusion.

Because the assignment requires attention to **elderly users, particularly those with visual impairments**, this test will also assess readability, clarity of navigation, button visibility, and ease of completing important tasks such as returns and issue reporting.

2. Objectives

The test will evaluate whether users can:

1. Find a product from the homepage.
2. Understand product information on the product page.
3. Add a product to cart and proceed to checkout.
4. Find and complete the return flow.
5. Find and complete the issue reporting flow.
6. Understand the labels, steps, buttons, and system feedback throughout the prototype.
7. Use the interface comfortably if they have reduced vision or are older adults.

3. Research Questions

1. Can users understand the overall site structure and navigation?
2. Can users complete core shopping tasks without help?
3. Can users locate after-sales services such as **Return** and **Report an Issue** easily?
4. Which screens, labels, or actions cause hesitation or errors?
5. Does the prototype support elderly users and users with visual impairments effectively?

4. Participants

- Target participants: 3–5 for the in-class pilot test, more for later independent tests

- Recommended profile:
 - General online shopping users
 - At least 1 participant aged 45+ if possible
 - At least 1 participant who reports mild difficulty with small text / visual clutter if possible

5. Method

Moderated task-based usability test

Roles

- **Facilitator:** welcomes participant, explains procedure, reads task scenarios, prompts think-aloud
- **Observer / Note-taker:** records time, errors, confusion points, comments, and behavior
- **Participant:** completes tasks using the prototype

6. Test Materials

- ShopCart prototype
- Consent form
- Pre-test questionnaire
- Post-test questionnaire
- Observation notes sheet / issues table
- Device used for testing: laptop or desktop

7. Test Environment

- Quiet tutorial room or study space
- Participant seated comfortably in front of a laptop
- Prototype opened and ready
- Screen zoom kept at default unless participant requests adjustment

8. Test Procedure

Introduction (2–3 min)

- Welcome participant
- Explain that **the prototype is being tested, not the participant**
- Ask participant to sign consent form
- Ask participant to complete the pre-test questionnaire

Test Session (10–15 min)

- Ask participant to think aloud while doing the tasks
- Facilitator reads one task at a time
- No help should be given unless the participant is completely stuck
- Observer records:
 - completion/success
 - time on task
 - navigation path
 - visible confusion
 - verbal comments
 - mistakes and recovery

Debrief (3–5 min)

- Ask follow-up questions
- Participant completes post-test questionnaire
- Thank participant

9. Task Scenarios

task scenarios.

Task 1: Find a product and view details

Scenario:

You want to buy a casual **black T-shirt** for everyday wear. Starting from the homepage, find a suitable product and open its product details page.

Success criteria:

- Participant reaches the relevant product page
- Participant can identify the product name, price, available colors, and size options

Task 2: Add the product to cart

Scenario:

You decide to buy the T-shirt. Add it to the cart.

Success criteria:

- Participant clicks **Add to Cart**
- Participant shows awareness that the item was added

Task 3: Review shopping cart and proceed to checkout

Scenario:

You want to review the items in your cart and continue to payment.

Success criteria:

- Participant reaches the purchase/cart page
- Participant understands the order summary
- Participant can identify the checkout action

Task 4: Return an item

Scenario:

After receiving your order, you realize the product is different from what you expected. You want to return it.

Success criteria:

- Participant reaches the Return page
- Participant proceeds in the return flow
- Participant can able to find return terms and conditions easily

Task 5: Report an issue with an order

Scenario:

You received an item and want to report a problem with it. Use the system to submit an issue report.

Success criteria:

- Participant reaches the Issue / Report an Issue page
- Participant selects the relevant item
- Participant understands the issue-reporting steps
- Participant submits or proceeds with issue reporting

Task 6: Use Assistive Tools – Voice Commands

Scenario:

You are shopping online, but you find it difficult to use the keyboard or mouse for a long time. You would like to use voice commands to help you complete a task. Starting from the homepage, use the voice command feature to search for a product and navigate to its product details page.

Success criteria:

- Participant can identify that a voice command option is available
- Participant understands how to activate or access the voice command feature
- Participant can use voice commands to search for a product or move to the next step
- Participant reaches the intended product details page with minimal confusion
-

10. Metrics Collected

Quantitative

- Task completion rate
- Time on task
- Number of errors
- Number of times participant asks for help
- Difficulty rating per task (1–5)

Qualitative

- Participant comments during think-aloud
- Confusion points
- Observed hesitation
- Navigation problems
- Visual accessibility concerns
- Suggestions for improvement

11. Observational Focus Areas

1. **Navigation discoverability**
Can users find it quicklycart, return, issue report?
2. **Label clarity**
“Return”, “Report an Issue”, “Continue”, “Submit Report” **Visual hierarchy**
3. **Readability for older adults**
Feedback and system status

12. Accessibility Focus

- text size too small
- low contrast
- clickable areas too small
- too much information on one screen
- icons without text labels
- important functions not prominent enough

13. Success Criteria for the Prototype

The prototype will be considered reasonably usable if:

- At least 80% of tasks are completed successfully
- Most participants can complete the main tasks without facilitator assistance

- Participants report moderate to high confidence
- Major usability issues are identifiable and actionable for revision

14. Ethical Considerations

- Participation is voluntary
- Participants may stop at any time
- No sensitive personal information is required
- Responses will be anonymized for analysis
- Testing data will be used only for this class project

Pre-test / Post-test questionnaires

Pre-test / Post-test questionnaires

Pre

ShopCart Prototype – Pre-test Questionnaire

Participant ID: _____

Date: _____

Age range:

- Under 18
- 18–24
- 25–34
- 35–44
- 45–54
- 55–64
- 65+

Section A: Background

1. How often do you shop online?
 - Never
 - Rarely
 - Sometimes
 - Often
 - Very often
2. Which devices do you usually use for online shopping?
 - Desktop / laptop 1
 - Tablet
 - Mobile phone
3. How confident are you when using online shopping websites/apps?
 - 1 Very unconfident
 - 2
 - 3 Neutral
 - 4 1
 - 5 Very confident
4. Have you ever returned a product purchased online?
 - Yes
 - No 1
5. Have you ever reported an issue or complaint about an online order?
 - Yes
 - No 1

Section B: Accessibility / Comfort

6. Do you usually have any difficulty reading text on websites?
 - Never
 - Rarely
 - Sometimes 1
 - Often

7. Do you use any accessibility support when browsing online?
- Glasses 1
 - Screen zoom
 - Screen reader
 - None
 - Other: _____
8. How comfortable are you reading small text on shopping websites?
- 1 Very uncomfortable
 - 2 1
 - 3 Neutral
 - 4
 - 5 Very comfortable
9. What do you usually expect to find easily on a shopping site?

Searchbar

10. What makes online shopping frustrating for you?

Lack of trustworthy reviews

Post

ShopCart Prototype – Post-test Questionnaire

Participant ID: _____

Date: _____

Section A: Overall Experience

1. Overall, how easy was the prototype to use?
 - 1 Very difficult
 - 2
 - 3 Neutral
 - 4 1
 - 5 Very easy
2. How easy was it to find products from the homepage?
 - 1 Very difficult
 - 2
 - 3 Neutral
 - 4 1
 - 5 Very easy
3. How easy was it to understand the product page?
 - 1 Very difficult
 - 2
 - 3 Neutral
 - 4 1
 - 5 Very easy
4. How easy was it to add a product to the cart and continue to checkout?
 - 1 Very difficult
 - 2
 - 3 Neutral 1
 - 4
 - 5 Very easy
5. How easy was it to find and use the Return function?
 - 1 Very difficult
 - 2
 - 3 Neutral 1
 - 4
 - 5 Very easy
6. How easy was it to find and use the Issue Reporting function?
 - 1 Very difficult
 - 2
 - 3 Neutral 1
 - 4
 - 5 Very easy

Section B: Interface Clarity

7. The labels and buttons were clear.
- 1 Strongly disagree
 - 2
 - 3 Neutral
 - 4 1
 - 5 Strongly agree
8. The layout of the pages was easy to follow.
- 1 Strongly disagree
 - 2
 - 3 Neutral
 - 4 1
 - 5 Strongly agree
9. The text size was readable.
- 1 Strongly disagree
 - 2
 - 3 Neutral
 - 4
 - 5 Strongly agree 1
10. Important actions and buttons were easy to notice.
- 1 Strongly disagree
 - 2
 - 3 Neutral 1
 - 4
 - 5 Strongly agree
11. I felt confident completing the tasks.
- 1 Strongly disagree
 - 2
 - 3 Neutral 1
 - 4
 - 5 Strongly agree

Section C: Open Questions

12. What part of the prototype was easiest to use?

The purchase button and features

13. What part of the prototype was most confusing?

The return policy feature

14. Was there any point where you felt lost or unsure what to do next?

Same as above

15. What changes would improve this prototype?

Improving clarity for the return policy and review feedback

16. Do you think this prototype would be easy for older adults or users with visual impairments to use? Why or why not?

Yes

17. Any additional comments?

More

Facilitator Script

Introduction Script

Hello, thank you for participating in our usability test.

We are testing our shopping prototype, not you. There are no right or wrong answers.

Please try to think aloud as you complete the tasks, so we can understand what you are thinking.

You may stop at any time if you feel uncomfortable.

We would like you to complete a short pre-test questionnaire first.

During the Test

- “Please tell me what you are thinking.”
- “What are you looking for now?”
- “What do you expect this button to do?”
- “Can you explain why you chose that?”

Don't say:

- “Click there”
- “That's wrong”
- “Use the menu”
- Any direct hints to the answer

Closing Script

Thank you for completing the test.

Please fill in this short post-test questionnaire.

Your feedback will help us improve the prototype.

Observation Sheet Template

Participant Observation Notes

Participant ID: _____

Observer: _____

Date: _____

Task	Success/Fail	Time	Errors	Asked for Help?	Notes / Quotes
Task 1 Find product					
Task 2 Add to cart					
Task 3 Checkout					
Task 4 Return item					
Task 5 Report issue					

Accessibility Notes

- Trouble reading text?
- Trouble identifying buttons?
- Trouble understanding page hierarchy?
- Trouble distinguishing icons?

final plan for submit

Our prototype link:

<https://www.figma.com/design/HGTPFbfQ6zh251Qc3GKJT9/E-Commerce-Website?node-id=1-529&t=Jz9hjzx9hbMjHiCC-1>

Usability Test Plan for ShopCart Prototype

1. Aim

The primary aim of this usability test is to evaluate the usability of the **ShopCart prototype**, an e-commerce interface that supports browsing, purchasing, returning products, reporting order issues, and accessing assistive features such as voice commands.

This usability test aims to identify any barriers, confusion points, or accessibility issues users may experience while interacting with the prototype. In particular, the evaluation focuses on whether users can complete common shopping and after-sales tasks efficiently and confidently, and whether the design supports **elderly users**, especially those with **visual impairments**. This reflects your original plan's purpose and accessibility focus.

2. Product Description

As part of this usability evaluation, we will test the **ShopCart prototype**. ShopCart is an e-commerce interface that allows users to:

- browse products from the homepage,
- view product details,
- add products to the cart,
- proceed to checkout,
- request returns,
- report issues with orders,
- and access assistive tools such as voice commands.

The product is currently in the prototype stage and is designed to support both general online shoppers and users with accessibility needs. The prototype includes enough functionality to test the key user tasks outlined in Section 6. This section matches the template's "Product Description" section on page 4.

3. Problem Statement Overview

The ShopCart prototype is designed to address key usability and accessibility issues commonly found in existing e-commerce platforms. Many current online shopping websites provide attractive interfaces and convenient purchasing functions, but they often do not sufficiently support users when problems arise after purchase. Refunds, return policies,

complaint procedures, and consumer rights information are often unclear or difficult to access, which may confuse and frustrate users.

In addition, many e-commerce websites are not fully accessible or user-friendly for elderly users or users with visual impairments. Dense layouts, small font sizes, low contrast, and complex navigation can make it difficult for these users to browse products, understand policies, and complete important tasks.

This usability test therefore focuses on evaluating whether the ShopCart prototype improves transparency, support, and accessibility by integrating shopping functions with clear return policies, complaint support, consumer-rights-related guidance, and a more accessible interface. This matches your plan's stated purpose and accessibility focus.

Product Problem Statements

Product Problem Statement	Priority
Are the terms, labels, and buttons on the website easy for users to understand?	1
Can users easily find and understand product details such as price, size, and colour options?	1
Can users add items to the cart and proceed to checkout without confusion?	1
Can users easily locate and complete the return process?	1
Can users easily locate and complete the issue reporting process?	1
Can users easily find and understand return terms, refund information, and complaint support?	1
Does the interface reduce confusion by presenting consumer support information clearly within the shopping process?	2
Is the interface accessible and usable for elderly users and users with visual impairments?	1
Do visual design choices such as font size, contrast, layout, and navigation support accessibility?	1
Can assistive features such as voice commands improve accessibility and task completion?	2

4. Participant Profile

This section follows the template's participant profile format on page 6.

We expect to test **3–5 participants** during the in-class pilot usability test, with more participants in later independent tests. Participants will be selected from general online shopping users where possible.

Participant Groups

Group	Number
General online shopping users	2–3
Older adults / participants aged 45+	1+
Participants with mild difficulty reading small text or dealing with visual clutter	1+

Participant Characteristics

Characteristic	Range / Description
Age	18–24, 25–44, 45+
Online shopping experience	Rarely to frequently shops online
Vision / accessibility needs	May include mild difficulty with small text, cluttered interfaces, or low contrast
Input preference	Standard mouse/keyboard use, with possible interest in assistive tools such as voice commands

5. Test Procedure

This section is based on the template's detailed procedure section on page 7.

1. Pre-test Questionnaire

Each participant will complete a short pre-test questionnaire before the usability evaluation session. This questionnaire will gather background information such as online shopping experience, comfort using shopping websites, and any accessibility-related needs.

2. Participant Greeting and Introduction

Each participant will be welcomed by the facilitator and made to feel comfortable and relaxed before the test begins.

3. Orientation and Consent

Participants will receive a verbal introduction explaining:

- the purpose of the usability test,
- that the **prototype is being tested, not the participant**,
- that they may stop at any time,
- and that their responses will only be used for this class project.

Participants will then sign a consent form before beginning the evaluation.

4. Usability Evaluation

The usability evaluation session will consist of a series of task-based activities using the ShopCart prototype.

The facilitator will:

- read each task scenario aloud,
- encourage the participant to think aloud,
- avoid leading the participant to the correct answer,
- and only provide help if the participant becomes completely stuck.

The observer will:

- record task completion or failure,
- time on task,
- navigation path,
- visible confusion,
- participant comments,
- and any mistakes or recovery strategies.

5. Participant Debriefing

After completing the tasks, participants will complete a post-test questionnaire and answer several follow-up questions about the usability, clarity, and accessibility of the prototype.

6. Task List

This section follows the template's structured task format on pages 8–9, where each task includes **Task**, **Starting State**, **Successful Completion Criteria**, **Benchmark**, **Script**.

Task 1

Task: Find a product and view details

Starting State: Homepage of the ShopCart prototype

Successful Completion Criteria:

- Participant reaches the relevant product page
- Participant identifies the product name, price, available colors, and size options

Benchmark: 2 minutes

Script (Task Scenario):

You want to buy a casual black T-shirt for everyday wear. Starting from the homepage, find a suitable product and open its product details page.

Task 2

Task: Add a product to the cart

Starting State: Product details page

Successful Completion Criteria:

- Participant clicks “Add to Cart”
- Participant shows awareness that the item was added

Benchmark: 1 minute

Script (Task Scenario):

You decide to buy the T-shirt. Add it to the cart.

Task 3

Task: Review cart and proceed to checkout

Starting State: Product page or cart access point

Successful Completion Criteria:

- Participant reaches the purchase/cart page
- Participant understands the order summary
- Participant identifies the checkout action

Benchmark: 2 minutes

Script (Task Scenario):

You want to review the items in your cart and continue to payment.

Task 4

Task: Return an item

Starting State: Homepage or account/order support access point

Successful Completion Criteria:

- Participant reaches the Return page
- Participant proceeds in the return flow
- Participant can find return terms and conditions easily

Benchmark: 2 minutes

Script (Task Scenario):

After receiving your order, you realize the product is different from what you expected. You want to return it.

Task 5

Task: Report an issue with an order

Starting State: Homepage or support access point

Successful Completion Criteria:

- Participant reaches the Issue / Report an Issue page
- Participant selects the relevant item
- Participant understands the issue-reporting steps
- Participant submits or proceeds with issue reporting

Benchmark: 2 minutes

Script (Task Scenario):

You received an item and want to report a problem with it. Use the system to submit an issue report.

Task 6

Task: Use assistive tools – voice commands

Starting State: Homepage of the ShopCart prototype

Successful Completion Criteria:

- Participant can identify that a voice command option is available
- Participant understands how to access the voice command feature
- Participant uses voice commands to search for a product or move to the next step
- Participant reaches the intended product details page with minimal confusion

Benchmark: 2 minutes

Script (Task Scenario):

You are shopping online, but you find it difficult to use the keyboard or mouse for a long time. You would like to use voice commands to help you complete a task.

Starting from the homepage, use the voice command feature to search for a product and navigate to its product details page.

7. Test Environment / Equipment

This section aligns with the template's "Test Environment/Application" section on page 10.

The usability testing session will be conducted in a quiet tutorial room or study space.

The following equipment will be used:

1. A laptop or desktop computer running the ShopCart prototype
2. Consent form
3. Pre-test questionnaire

4. Post-test questionnaire
5. Observation notes sheet / issues table

Participants will be seated comfortably in front of the computer. The prototype will be opened and ready before the test begins.

8. Test Monitor Role

This section follows the template's "Test Monitor Role" guidance on page 11.

The test monitor will act as the **facilitator** during the evaluation session. The facilitator will remain in the room with the participant and guide the session in a neutral manner.

The facilitator will:

- read each task scenario aloud,
- encourage the participant to think aloud,
- avoid leading the participant toward the correct answer,
- and only provide help when the participant is at serious risk of being unable to continue.

An observer / note-taker will also be present to make observations about:

- user confusion,
- navigation errors,
- hesitation,
- participant comments,
- and any accessibility concerns not immediately captured through timing or success/failure data.

9. Evaluation Measures / Data to be Collected

This section matches the template's "Evaluation Measures / Data to be collected" table on page 12.

Performance Measures

- Number and percentage of tasks completed successfully
- Number of tasks completed with and without assistance
- Time to complete individual tasks
- Total time to complete all tasks
- Number of errors encountered during each task
- Number of times the participant asks for help
- Difficulty rating per task (1–5)

Behavioural / Qualitative Measures

- Participant comments during think-aloud
- Confusion points
- Observed hesitation
- Navigation problems
- Visual accessibility concerns
- Participant perception of the interface
- Suggestions for improvement
- Relevant positive and negative quotes

Accessibility-Specific Measures

- Whether text size is too small
- Whether contrast is too low
- Whether clickable areas are too small
- Whether screens feel too crowded
- Whether icons without labels create confusion
- Whether important functions are prominent enough
- Whether the voice command tool is discoverable and understandable

10. Report Contents

This section follows the template's "Report Contents" section on page 13.

A formal report will be produced summarising:

- the results of the usability evaluation,
- task completion outcomes,
- major usability problems identified,
- accessibility-related findings,
- and recommendations for improving the ShopCart prototype.

The findings will later be used to support prototype revision and the final presentation for the assignment.

11. Ethical Considerations

This section is already in your original plan and should stay.

- Participation is voluntary
- Participants may stop at any time
- No sensitive personal information is required

- Responses will be anonymized for analysis
- Testing data will be used only for this class project

Consent form for participation in ShopCart User Research

Please read the following information carefully.

Student Kapil Vinayak is conducting today's research on behalf of The University of New South Wales. The research is being used as a class assessment exercise for a course Human Computer Interaction (COMP3511) at The University of New South Wales. The topic is concerned with the design of *e-commerce web application*.

As a participant in the session, we will ask you to help us understand how you work and your wants and needs for e-commerce web application. The observations and information we collect today will help us to build improvements into the designs and will be used as part of our University assignment work that help us to learn about involving users in the design process. If you have any questions, we encourage you to ask them.

As your participation is entirely voluntary, you may choose to withdraw from the session at any time.

If you have any further queries you can contact Student Kapil Vinayak on Kapil10@gmail.com.

Your privacy, observers & recording of data

Please be assured that the information and opinions you provide today will be used only for research purposes.

We will record information about how you might use and interact with the site (including on videotape, audiotape and questionnaire). This will help us see how to make the design more useful and easier to use. For any and all uses of this recorded data, we undertake to guarantee your anonymity. Your name will appear only on this form, which will be filed separately, and is kept only to verify your consent to participation in this session. The form will be destroyed after the completion of the Human Computer Interaction course.

The data recorded may be used for:

- Analysis, to review interactions after the session
- Documenting the findings of the session
- Providing feedback to the project team about the design and how well it does/does not meet peoples' needs or communicating the same to other stakeholders within the organisation sponsoring the research
- Teaching or training purposes by group members but only for demonstrating the process of a typical research session (eg. how to run this type of session).

- Notes and questionnaires may be passed on to teaching staff of the University for course assessment purposes. If you sign this sheet, you give your permission to have notes recorded and for the notes to be passed on to course teaching staff.
- The notes will not be transferred to anyone else unless your consent has been obtained.

Please continue over...

Non-disclosure

By participating in this session, you may be privy to information that the course regards as confidential, and must accept that the participant is the sole and exclusive owner of this confidential information.

You must not at any time after the session disclose to any other person or use, adopt, be a party to, aid or abet any disclosure, use, adoption or employment of this confidential information either directly or indirectly for any purpose.

If you have read the above information carefully and understand and consent to its contents, please complete and sign the following statement.

I,

(Participant's Name – please print)

have read the above passage carefully and understand its contents. I voluntarily agree to participate in the e-commerce web application. evaluation conducted by Jiajun Tang.

(Participant's Signature)

(Date)

(Researcher's
Signature)

(Date)

ShopCart Prototype – Pre-test Questionnaire

Participant ID: _____

Date: _____

Age range:

- Under 18
- 18–24
- 25–34
- 35–44
- 45–54
- 55–64
- 65+

Section A: Background

1. How often do you shop online?
 - Never
 - Rarely
 - Sometimes
 - Often
 - Very often
2. Which devices do you usually use for online shopping?
 - Desktop / laptop 1
 - Tablet
 - Mobile phone
3. How confident are you when using online shopping websites/apps?
 - 1 Very unconfident
 - 2
 - 3 Neutral
 - 4 1
 - 5 Very confident
4. Have you ever returned a product purchased online?
 - Yes
 - No 1
5. Have you ever reported an issue or complaint about an online order?
 - Yes
 - No 1

Section B: Accessibility / Comfort

6. Do you usually have any difficulty reading text on websites?
 - Never
 - Rarely
 - Sometimes 1
 - Often
7. Do you use any accessibility support when browsing online?

- Glasses 1
 - Screen zoom
 - Screen reader
 - None
 - Other: _____
8. How comfortable are you reading small text on shopping websites?
- 1 Very uncomfortable
 - 2 1
 - 3 Neutral
 - 4
 - 5 Very comfortable
9. What do you usually expect to find easily on a shopping site?

Searchbar

10. What makes online shopping frustrating for you?

Lack of trustworthy reviews

ShopCart Prototype – Post-test Questionnaire

Participant ID: _____

Date: _____

Section A: Overall Experience

1. Overall, how easy was the prototype to use?
 - 1 Very difficult
 - 2
 - 3 Neutral
 - 4 1
 - 5 Very easy
2. How easy was it to find products from the homepage?
 - 1 Very difficult
 - 2
 - 3 Neutral
 - 4 1
 - 5 Very easy
3. How easy was it to understand the product page?
 - 1 Very difficult
 - 2
 - 3 Neutral
 - 4 1
 - 5 Very easy
4. How easy was it to add a product to the cart and continue to checkout?
 - 1 Very difficult
 - 2
 - 3 Neutral 1
 - 4
 - 5 Very easy
5. How easy was it to find and use the Return function?
 - 1 Very difficult
 - 2
 - 3 Neutral 1
 - 4
 - 5 Very easy
6. How easy was it to find and use the Issue Reporting function?
 - 1 Very difficult
 - 2
 - 3 Neutral 1
 - 4
 - 5 Very easy

Section B: Interface Clarity

7. The labels and buttons were clear.

- 1 Strongly disagree
 - 2
 - 3 Neutral
 - 4 1
 - 5 Strongly agree
8. The layout of the pages was easy to follow.
- 1 Strongly disagree
 - 2
 - 3 Neutral
 - 4 1
 - 5 Strongly agree
9. The text size was readable.
- 1 Strongly disagree
 - 2
 - 3 Neutral
 - 4
 - 5 Strongly agree 1
10. Important actions and buttons were easy to notice.
- 1 Strongly disagree
 - 2
 - 3 Neutral 1
 - 4
 - 5 Strongly agree
11. I felt confident completing the tasks.
- 1 Strongly disagree
 - 2
 - 3 Neutral 1
 - 4
 - 5 Strongly agree

Section C: Open Questions

12. What part of the prototype was easiest to use?

The purchase button and features

13. What part of the prototype was most confusing?

The return policy feature

14. Was there any point where you felt lost or unsure what to do next?

Same as above

15. What changes would improve this prototype?

Improving clarity for the return policy and review feedback

16. Do you think this prototype would be easy for older adults or users with visual impairments to use? Why or why not?

Yes

17. Any additional comments?

Observation Notes

Project: ShopCart Prototype Usability Test
Participant ID: P01
Date: [Insert Date]
Observer: [Insert Name]
Test Type: Moderated usability test

Overall Observation

The participant was able to complete the main tasks and reported that the prototype was generally easy to use. Finding products from the homepage, understanding the product details page, and using the purchase-related functions appeared to be relatively smooth. The participant gave positive feedback on the layout, labels, buttons, and especially text readability, suggesting that the interface performed reasonably well in terms of basic usability and visual clarity.

However, the participant experienced noticeable confusion when using the **Return** function. In the post-test questionnaire, they identified the **return policy feature** as the most confusing part of the prototype and stated that this was the point where they felt unsure about what to do next. The **Issue Reporting** feature was usable, but only at a moderate level of clarity. Overall, the prototype performed better in the **shopping and purchase flow** than in the **after-sales support flow**, especially in relation to return policy guidance.

Task-by-Task Observation Notes

Task	User Actions	Navigation Difficulty	Outcome	Notes
Task 1: Find a product and view details	The participant was able to browse from the homepage and reach the black T-shirt product details page.	Mild difficulty	Successful	Product discovery was generally easy, but the product detail entry point could be clearer.

Task 2: Add a product to the cart	The participant identified the purchase button and added the product to the cart successfully.	Easy	Successful	The participant later reported that the purchase button and purchase-related features were the easiest parts of the prototype.
Task 3: Review cart and proceed to checkout	The participant was able to access the cart / purchase page and understand the order summary.	Moderate	Successful	The task could be completed, but the transition from cart to checkout may still need clearer guidance.
Task 4: Return an item	The participant was able to reach the Return section, but experienced confusion when trying to understand the return policy and the next step in the process.	Difficult	Partially smooth / completed with confusion	This was the most problematic task. The participant reported feeling lost at this stage.
Task 5: Report an issue with an order	The participant was able to find and use the Issue Reporting function.	Moderate	Successful	The feature was usable, but the participant's ratings suggest that the process was not fully intuitive.

Detailed Notes by Task

Task 1: Find a product and view details

The participant was able to locate the target product from the homepage and open the black T-shirt product details page. Based on the post-test rating, finding products from the

homepage was relatively easy. This suggests that the homepage layout, product cards, and product browsing flow were reasonably understandable.

Interpretation: Navigation was effective overall, although the clickable entry to the product details page could be made more obvious.

Task 2: Add a product to the cart

The participant was able to identify the purchase button and complete the add-to-cart action without major difficulty. In the post-test responses, the participant explicitly stated that the **purchase button and related features** were the easiest part of the prototype to use.

Interpretation: The main call-to-action design was effective and aligned with user expectations.

Task 3: Review cart and proceed to checkout

The participant was able to access the cart / purchase page and review the order summary. However, the related ratings were only moderate rather than very high, suggesting that while the flow was usable, it may not have felt completely seamless.

Interpretation: The shopping cart and checkout flow was functional, but the next-step guidance and information hierarchy could be improved.

Task 4: Return an item

This was the most difficult task for the participant. The participant explicitly reported that the **return policy feature** was the most confusing part of the prototype, and also stated that this was the point where they felt unsure about what to do next.

Interpretation: The Return section has significant usability issues, likely involving one or more of the following:

- unclear return policy wording,
- weak visibility of return-related information,
- insufficient step-by-step guidance,
- unclear return terms and conditions,
- or a lack of feedback about what to do next.

Task 5: Report an issue with an order

The participant was able to complete the Issue Reporting task, but the feature received only moderate ease-of-use ratings. This indicates that the participant could use it, but did not find it especially clear or intuitive.

Interpretation: The feature is functional, but could benefit from clearer wording, stronger visual guidance, and more confidence-building feedback.

Post-test Questionnaire Summary

Category	Summary
Overall usability	Generally easy to use
Finding products from homepage	Relatively easy
Understanding the product page	Relatively easy
Add to cart and continue to checkout	Moderate
Return function	Moderate, with noticeable confusion
Issue Reporting function	Moderate, usable but not highly intuitive
Labels and buttons	Fairly clear
Page layout	Easy enough to follow
Text size	Very readable
Visibility of important actions	Moderate
Confidence in completing tasks	Moderate

Key Open-Ended Responses

- **Easiest part:** The purchase button and purchase-related features

- **Most confusing part:** The return policy feature
 - **Point of feeling lost:** Same as above, in the Return section
 - **Suggested improvements:** Improve the clarity of the return policy and review feedback
 - **Suitability for older adults / visually impaired users:** Yes
-

Issues Table

Issue ID	Related Task	Problem Observed	Severity	Evidence	Impact	Suggested Fix
1	Task 4	The return policy feature was unclear and confusing to the participant.	3	The participant identified the return policy feature as the most confusing part of the prototype.	Reduces usability of the Return flow and makes after-sales support harder to use.	Make return policy information more visible and rewrite it using clearer, simpler wording.
2	Task 4	The participant felt lost during the Return process and was unsure what to do next.	3	The participant reported that this was the point where they felt unsure about the next step.	Breaks flow continuity and reduces task confidence.	Add clearer step-by-step guidance, stronger visual hierarchy, and a more obvious next action.

3	Task 5	The Issue Reporting process was usable but not fully intuitive.	2	The Issue Reporting feature received only moderate ease-of-use ratings.	The task can be completed, but with some cognitive effort.	Simplify labels and provide clearer guidance throughout the reporting process.
4	Task 3	The cart-to-check out flow was functional but only moderately clear.	2	Add-to-cart and checkout received moderate rather than very high ratings.	May reduce smoothness in the purchase flow.	Strengthen the visibility of the checkout action and improve the structure of the order summary.
5	Global	Important actions and buttons were not always immediately noticeable.	2	“Important actions and buttons were easy to notice” received only a moderate rating.	Can lead to hesitation and slower navigation.	Increase contrast, button prominence, and visual consistency for key actions.
6	Global	The participant's confidence in completing the tasks was only moderate.	2	“I felt confident completing the tasks” received a moderate rating.	Suggests that some flows still require clearer guidance.	Improve feedback messages, flow consistency, and action visibility across the interface.

7	Global (Positive Finding)	Text readability was strong.	0	The participant gave a very positive rating for text size readability.	Supports usability for older adults and users with visual impairments.	Retain the current text size and readability approach.
8	Task 2 (Positive Finding)	Purchase-related actions were clear and easy to use.	0	The participant explicitly said that the purchase button and purchase-related features were the easiest part to use.	Indicates that the main shopping flow is effective.	Retain the current CTA design style and apply similar clarity to after-sales functions.

Short Summary for Report

Based on this usability test, the participant was able to complete all major tasks in the ShopCart prototype, and the interface was generally perceived as easy to use. The product browsing flow, product details page, and purchase-related actions performed well, with the purchase button identified as the easiest feature to use. The prototype also showed strengths in readability, particularly in text size, which supports accessibility for older adults and users with visual impairments.

However, the Return section presented the most significant usability issues. The participant found the return policy feature confusing and reported feeling unsure about what to do next during this part of the process. The Issue Reporting feature was usable but only moderately intuitive. Overall, the findings suggest that the prototype is stronger in the shopping and purchase flow than in the after-sales support flow, and that future revisions should prioritise clearer return policy communication, stronger guidance in the Return flow, and improved visibility of important actions.
